STA 130A: Mathematical Statistics: Brief Course

Midterm exam 1

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Name: _____ **Student ID:** _____

Instructions: This midterm exam is **closed-book**, except for the permitted note sheet described below. You may bring a pen or pencil, one letter-sized sheet of *hand-written* notes (both sides), and a *non-graphing* calculator. No other materials (e.g., textbooks) are allowed. You have 50 minutes for the exam. The **total score is 120 points**, with *up to 10 bonus points available*. Once you receive this exam, please confirm that you have all 8 pages, including the standard normal table in the appendix.

- Make sure to clearly write your name and student ID above.
- Present answers succinctly, but include all relevant steps for full credit. Partial credit is possible only if your reasoning is clearly shown and traceable by the grader.
- If necessary, round numerical answers to three decimal places. Exact expressions, including fractions and counting expressions, are acceptable.
- Bonus problems can be more challenging; consider attempting them after you finish the main problems.

Problem	**Score**
Problem 1	
Problem 2	
Problem 3	
Problem 4	
Bonus	
Total	

Problem 1 (40 points in total). Warm-up

Treat each item separately. Most items require only one or two lines of calculation or justification.

(a) (5 points) Independent trials each have success probability $1/4$. Let N be the trial number of the first success. Calculate $P(N = 3)$.

(b) (5 points) Events A and B are disjoint, with $P(A) = 1/4$ and $P(B) = 1/3$. Can they be independent? Give a reason in one sentence.

(c) (5 points) How many three-digit codes can be formed using the digits $1, 2, \ldots, 6$, if digits cannot repeat and the final digit must be 6?

(d) (5 points) Suppose $\mathbb{E}[X] = 2$ and $\mathbb{E}[Y - X] = 3$. Calculate $\mathbb{E}[Y]$.

(e) (5 points) Let $I_1, \ldots, I_6$ be independent Bernoulli$(1/4)$ random variables. Identify the distribution of $6 - \sum_{i=1}^{6} I_i$ by giving **either** its name and parameters **or** its PMF.

(f) (5 points) Can a CDF F satisfy $F(0) = 0.7$ and $F(1) = 0.6$? Explain the reason in one sentence.

(g) (5 points) Suppose $P(C) = 3/5$, $P(D) = 1/2$, and $P(C \cup D) = 9/10$. Calculate $P(C \cap D^c)$.

(h) (5 points) Suppose $P(A) = 1/3$, $P(B \mid A) = 3/4$, and $P(B \mid A^c) = 3/8$. Calculate $P(A \mid B)$.

Problem 2 (25 points in total). Joint distributions and moments

The joint PMF of X and Y is

$p_{X,Y}(x,y)$	$y=0$	$y=1$
$x=0$	3/16	1/16
$x=1$	2/16	6/16
$x=2$	3/16	1/16

and $p_{X,Y}(x,y)=0$ for all other pairs.

(a) (7 points) Write the marginal PMF $p_X(x)$.

(b) (6 points) Determine whether X and Y are independent, and justify your answer.

(c) (6 points) Calculate $\mathbb{E}[X^2 \mid Y=1]$.

(d) (6 points) Calculate $\mathrm{Var}(X+Y)$.

Problem 3 (25 points in total). Densities and functions of random variables

Let X have PDF

$$f_X(x) = \begin{cases} c(1+x), & 0 < x < 2, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant.

(a) (6 points) Find the value of c.

(b) (10 points) Define

$$Q = \begin{cases} 0, & X \leq 1, \\ X - 1, & X > 1. \end{cases}$$

Calculate $\mathbb{E}[Q]$, leaving your answer **in terms of** c.

(c) (9 points) Separately, let $Y \sim N(10, 2^2)$. Calculate $P(8 \leq Y \leq 13)$ using the standard normal table in the appendix.

Problem 4 (30 points in total). Assigning tasks

Three distinct tasks must be assigned to three different workers from a pool of two experienced workers and four trainees, one task per worker.

(a) (10 points) How many distinct task-to-worker assignments include exactly one experienced worker? Use permutations and/or combinations, and explain the factors in your count.

For parts (b)–(c): The supervisor first chooses how many experienced workers to assign to the three tasks. Let X denote this number of experienced workers, with the following PMF:

x	0	1	2
$P(X = x)$	1/4	1/2	1/4

All three tasks start at time 0 in parallel, each by its assigned worker. For any fixed assignment,

- each experienced worker finishes their task in exactly 1/4 hour;
- the trainees' completion times are independent random variables, uniformly distributed on $(0, 2)$ hours.

Let Y be the number of tasks completed by time 1 hour.

(b) (10 points) Calculate the expected number of tasks completed by time 1 hour. Show your reasoning.

(c) (10 points) Let T be the time, in hours, at which the last of the three tasks is completed.

(i) Calculate $P(X = 2 \mid T \leq 1/2)$.

(ii) Interpret your answer in one sentence, comparing it with $P(X = 2)$.

Optional bonuses (5 points each)

Complete the required problems first. A brief, well-supported argument is sufficient.

Bonus 1 (5 points). Integer values and variance

An integer-valued random variable X has $\mathbb{E}[X] = 3/2$ and finite variance.

(i) What is the smallest possible value of $\mathrm{Var}(X)$?
(ii) Write a PMF attaining this minimum.
(iii) Give a brief argument that no smaller value is possible.

Bonus 2 (5 points). Three equally likely labels

Suppose we can toss a fair coin independently up to three times. We want to design a rule that always generates exactly one label from $\{A, B, C\}$, using only the observed tosses and no other randomness.

(i) Can such a rule give each label the same probability $1/3$?
(ii) If yes, give such a rule and verify its probabilities; if no, prove that no such rule exists.

Appendix. Standard normal table

For $Z \sim N(0,1)$, this table gives $\Phi(z) = P(Z \leq z)$. The row gives the units and tenths of z, and the column gives the hundredths. For negative z, use $\Phi(-z) = 1 - \Phi(z)$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990